Hospital Charges– Data Analysis

Business Objective

Hospitals generate extensive data related to patient discharges, diagnoses (DRGs), covered charges, total payments, and Medicare reimbursements. However, healthcare administrators and policymakers lack clear, actionable insights into cost variations, payment inefficiencies, and drivers of high hospital charges across states, hospitals, and diagnoses.

This project aims to conduct an end-to-end analysis of the Hospital Charges dataset to:

1. Identify key drivers influencing hospital charges and Medicare payments.
2. Understand variations in charges and payments across states, hospitals, and DRGs.
3. Evaluate gaps between covered charges, total payments, and Medicare reimbursements.
4. Assess the impact of discharge volume on hospital revenue and payments.
5. Provide data-driven recommendations to improve cost efficiency, transparency, and healthcare policy decisions.

Problem Statement

Healthcare organizations and policymakers seek to understand the factors contributing to high hospital charges and payment disparities. Despite having detailed hospital billing and Medicare payment data, decision-makers lack clarity on:

- Why charges vary significantly across hospitals and states
- Which diagnoses drive the highest costs
- How Medicare reimbursements compare to billed charges
- Where inefficiencies and cost escalation occur

The goal is to analyze historical hospital charge data to identify trends, uncover inefficiencies, and provide actionable insights to support cost control, policy formulation, and healthcare optimization.

Phase 1: Business Understanding

Business Questions

- Which hospitals and states have the highest average covered charges?
- How do hospital charges vary across different DRGs?
- What is the gap between covered charges and Medicare payments?
- How does total discharge volume impact total payments?
- Which hospitals show higher Medicare reimbursement efficiency?

Phase 2: Data Understanding

Data Exploration

Q1. Steps Taken During Data Exploration

The following steps were performed during data exploration:

1. Reviewed dataset structure and attribute definitions.
2. Identified categorical and numerical variables.
3. Checked for missing, null, and inconsistent values.
4. Verified data types and value ranges.
5. Analyzed distribution of covered charges, total payments, and Medicare payments.
6. Examined hospital-level and state-level charge patterns.
7. Reviewed DRG-wise cost and payment distribution.
8. Identified outliers in charges and payments.
9. Performed initial aggregations and pivot table analysis.
10. Analyzed variability in discharge volumes across hospitals.

Q2. Insights Gained During Data Exploration

The following insights were identified:

- Significant variation exists in hospital charges across states and hospitals.
- Average covered charges are substantially higher than Medicare payments.
- Certain DRGs consistently exhibit higher costs.
- High discharge volume does not always translate to higher efficiency.
- Medicare payments remain relatively standardized compared to billed charges.
- Some hospitals show unusually high charge-to-payment gaps.
- Urban and specialized hospitals tend to report higher charges.
- State-level policies appear to influence average payment patterns.

Q3. Calculated Columns Created

The following calculated columns were created:

1. Charge-to-Payment Ratio
   Average Covered Charges ÷ Average Medicare Payments
2. Payment Gap Amount
   Average Covered Charges − Average Medicare Payments
3. Discharge Volume Category
   Low, Medium, High based on total discharges
4. Hospital Cost Category
   Low Cost / Medium Cost / High Cost
5. State-Level Cost Index
   Hospital charges compared to state average
6. DRG Cost Classification
   High-cost vs Standard-cost DRGs

Q4. KPIs Created and Analysis

Key Performance Indicators (KPIs)

- Total Hospitals Analyzed
- Total Discharges
- Average Covered Charges
- Average Total Payments
- Average Medicare Payments
- Charge-to-Payment Ratio
- Payment Efficiency Index

Insights from KPIs

- Covered charges are multiple times higher than Medicare reimbursements.
- Hospitals with moderate discharge volumes often show better efficiency.
- Certain states consistently report higher charge-to-payment ratios.
- DRG type strongly influences cost and payment behavior.
- Payment efficiency varies significantly by hospital.

Phase 3: Data Preparation

Data Cleaning

1. Handling Dimension and Measure Names

- Standardized column naming conventions.
- Removed special characters and unnecessary symbols.
- Renamed ambiguous fields for clarity and consistency.

2. Handling Missing Values

- Identified and handled missing or null charge/payment values.
- Removed records with incomplete financial information.
- Ensured DRG and hospital identifiers were complete.

3. Data Type Formatting

- Converted categorical fields (State, DRG, Hospital) to text.
- Ensured numerical fields were formatted correctly.
- Verified monetary values were consistent and positive.

Feature Engineering

- Created discharge volume categories.
- Derived charge-to-payment and payment gap metrics.
- Created state-level benchmarking indicators.
- Classified hospitals by cost intensity.
- Grouped DRGs into cost categories.

Data Integration and Formatting

- Ensured consistency between hospital, DRG, and state data.
- Removed duplicate hospital-DRG records.
- Standardized state abbreviations.
- Prepared dataset for analysis and visualization.

Phase 3.1: Data Exploration and Analysis

Analysis Performed

- State-wise charge and payment comparison.
- Hospital-level efficiency analysis.
- DRG-wise cost and reimbursement evaluation.
- Discharge volume vs total payment analysis.
- Identification of high-cost and inefficient hospitals.
- Gap analysis between charges and Medicare payments.
- Outlier detection in hospital billing patterns.

Key Findings and Recommendations

Key Findings

- Hospital charges vary widely across states and providers.
- Medicare payments cover only a fraction of billed charges.
- Certain DRGs consistently drive higher hospital costs.
- Higher discharge volumes do not guarantee better payment efficiency.
- Some hospitals demonstrate significantly higher billing inefficiencies.

Recommendations

1. Monitor and regulate high charge-to-payment ratio hospitals.
2. Promote transparency in hospital billing practices.
3. Encourage efficiency-based reimbursement models.
4. Focus cost-control efforts on high-cost DRGs.
5. Use state-level benchmarks to identify pricing outliers.